

Group Project Presentation
MIS 776 Fall 2025
Dr. Michael Lee

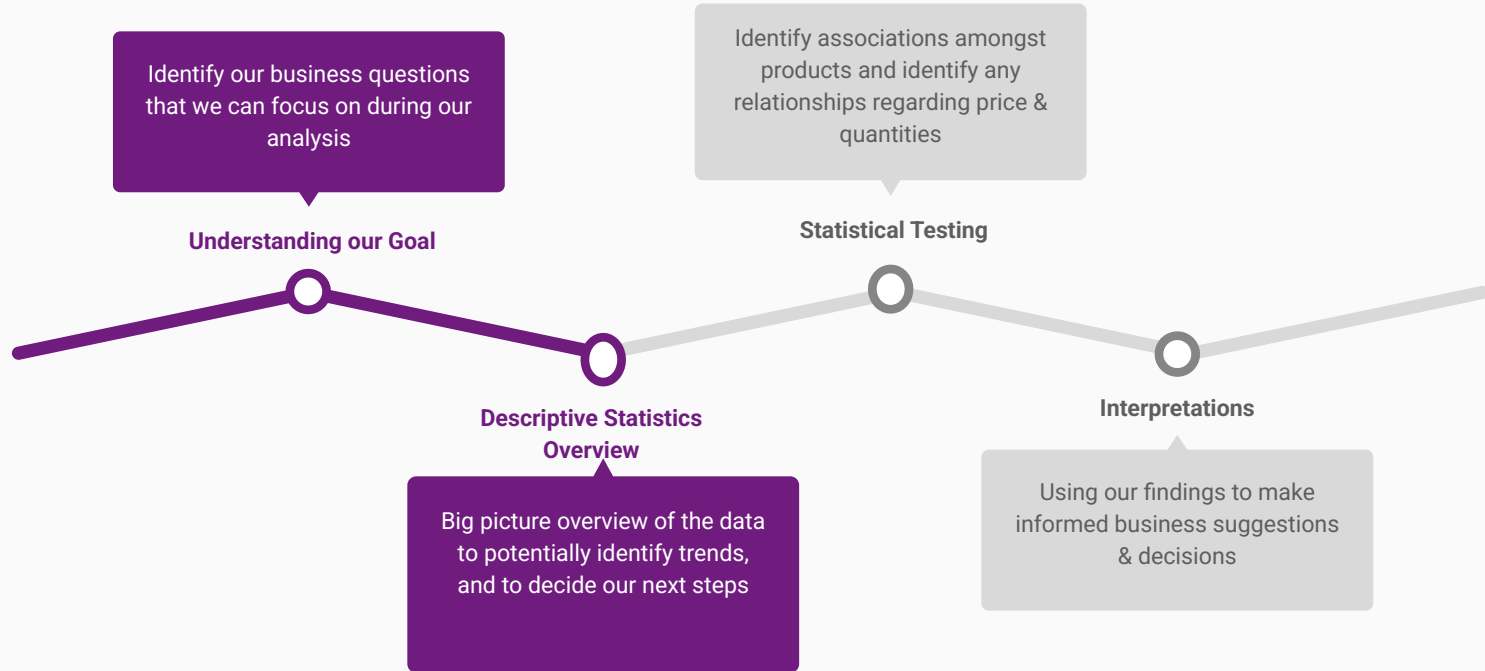
Haley, Jessie, Kevely, Saeed



Introduction

- **onlineRetail2** dataset contains data pertaining to customer purchase behaviors including products, prices, and timestamps
- Question 1 - Can we estimate the probability of future purchases by identifying associations between products?
- Question 2 - Are there meaningful insights that emerge from differences in product prices and quantities?

Overview of the Analysis





Dataset overview

Why we're
interested

Descriptive
statistics



Online Retail II

Donated on 9/20/2019

A real online retail transaction data set of two years.

Dataset Characteristics

Multivariate, Sequential, Time-Series,
Text

Subject Area

Business

Associated Tasks

Classification, Regression, Clustering

Feature Type

Integer, Real

Instances

1067371

Features

-

Dataset Information

Additional Information

This Online Retail II data set contains all the transactions occurring for a UK-based and registered, non-store online retail between 01/12/2009 and 09/12/2011. The company mainly sells unique all-occasion gift-ware. Many customers of the company are wholesalers.

Has Missing Values?

Yes

Rows and columns: (397925, 10)

Summary Statistics

	Quantity	Price
count	397925.000000	397925.000000
mean	13.021793	3.116212
std	180.419984	22.096773
min	1.000000	0.000000
25%	2.000000	1.250000
50%	6.000000	1.950000
75%	12.000000	3.750000
max	80995.000000	8142.750000

Greatest Quantity Items:

	Description	Quantity
0	PAPER CRAFT , LITTLE BIRDIE	80995
1	MEDIUM CERAMIC TOP STORAGE JAR	77916
2	WORLD WAR 2 GLIDERS ASSTD DESIGNS	54415
3	JUMBO BAG RED RETROSPOT	46181
4	WHITE HANGING HEART T-LIGHT HOLDER	36725
5	ASSORTED COLOUR BIRD ORNAMENT	35362
6	PACK OF 72 RETROSPOT CAKE CASES	33693
7	POPCORN HOLDER	30931
8	RABBIT NIGHT LIGHT	27202
9	MINI PAINT SET VINTAGE	26076

Average Items Per Invoice: 26.45360809197713



Rows with negative Quantity: 8905

First Rows

	Invoice	StockCode	Description	Quantity
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6
1	536365	71053	WHITE METAL LANTERN	6
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART	6

	InvoiceDate	Price	Customer ID	Country
0	12/1/10	2.55	17850.0	United Kingdom
1	12/1/10	3.39	17850.0	United Kingdom
2	12/1/10	2.75	17850.0	United Kingdom
3	12/1/10	3.39	17850.0	United Kingdom
4	12/1/10	3.39	17850.0	United Kingdom

Total Revenues

Country		Unspecified	2667.070
United Kingdom	7308391.554	United Arab Emirates	1902.280
Netherlands	285446.340	Lebanon	1693.880
EIRE	265545.900	Lithuania	1661.060
Germany	228867.140	European Community	1300.250
France	209042.050	Brazil	1143.600
Australia	138521.310	RSA	1002.310
Spain	61577.110	Czech Republic	826.740
Switzerland	56443.950	Bahrain	548.400
Belgium	41196.340	Saudi Arabia	145.920
Sweden	38378.330		
Japan	37416.370		
Norway	36165.440		
Portugal	33439.890		
Finland	22546.080		
Singapore	21279.290		
Channel Islands	20450.440		
Denmark	18955.340		
Italy	17483.240		
Cyprus	13590.380		
Austria	10198.680		
Poland	7334.650		
Israel	7221.690		
Greece	4760.520		
Iceland	4310.000		
Canada	3666.380		
USA	3580.390		
Malta	2725.590		

Sample of Items

Unique: 3866



Description	
EDWARDIAN PARASOL NATURAL	
CHRISTMAS LIGHTS 10 REINDEER	
50'S CHRISTMAS GIFT BAG LARGE	
FAIRY DREAMS INCENSE	
TRADITIONAL MODELLING CLAY	
BOX OF 6 MINI 50'S CRACKERS	
TREASURE TIN GYMKHANA DESIGN	
FOLDING UMBRELLA PINKWHITE POLKADOT	
SKULLS TAPE	
JUMBO BAG PAISLEY PARK	
CHERRY BLOSSOM DECORATIVE FLASK	
SET OF 3 HANGING OWLS OLLIE BEAK	
JUMBO BAG APPLES	
PAPER CHAIN KIT RETROSPOT	
PINK HEARTS LIGHT CHAIN	
RED RETROSPOT CHARLOTTE BAG	
REGENCY CAKESTAND 3 TIER	
BELLE JARDINIERE CUSHION COVER	
SET OF 3 BUTTERFLY COOKIE CUTTERS	
PING MICROWAVE APRON	



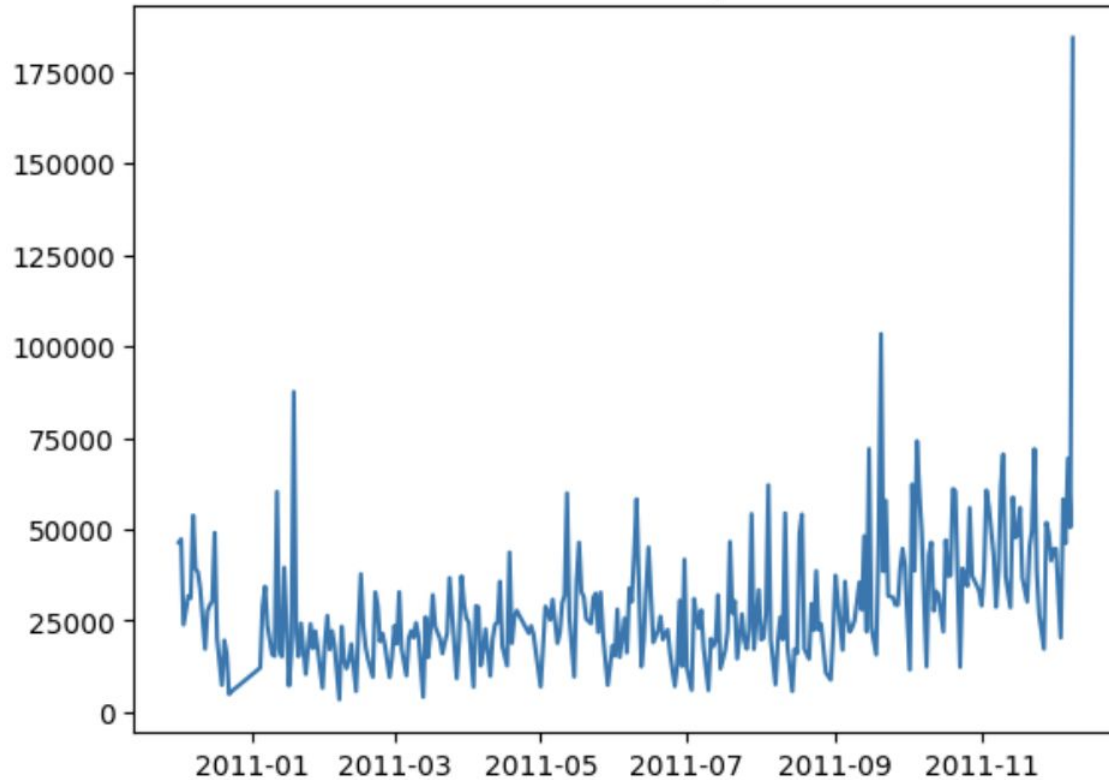
```
data['date_revenue'] = data['Price'] * data['Quantity']
data['InvoiceDate'] = pd.to_datetime(data['InvoiceDate'])
date_revenue = data.groupby('InvoiceDate')['date_revenue'].sum()
date_revenue = pd.DataFrame(date_revenue)
print(date_revenue)
plt.plot(date_revenue['date_revenue'])
plt.title("Total Revenues By Date")
plt.show()
```

✓ 0.2s

InvoiceDate	date_revenue
2010-12-01	46376.49
2010-12-02	47316.53
2010-12-03	23921.71
2010-12-05	31771.60
2010-12-06	31215.64
...	...
2011-12-05	58202.21
2011-12-06	46144.04
2011-12-07	69354.21
2011-12-08	50519.41
2011-12-09	184367.28

[305 rows x 1 columns]

Total Revenues By Date



```
multiply_items = data.loc[data.index.repeat(data['Quantity'])]
multiply_items = multiply_items.drop('Quantity', axis=1)
print(multiply_items.head(50))
print("\nRows and columns expanding invoice items:", multiply_items.shape)
```

✓ 1.3s

	Invoice	StockCode	Description	InvoiceDate	Price	\
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	12/1/10	2.55	
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	12/1/10	2.55	
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	12/1/10	2.55	
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	12/1/10	2.55	
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	12/1/10	2.55	
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	12/1/10	2.55	
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	12/1/10	2.55	
1	536365	71053	WHITE METAL LANTERN	12/1/10	3.39	
1	536365	71053	WHITE METAL LANTERN	12/1/10	3.39	
1	536365	71053	WHITE METAL LANTERN	12/1/10	3.39	
1	536365	71053	WHITE METAL LANTERN	12/1/10	3.39	
1	536365	71053	WHITE METAL LANTERN	12/1/10	3.39	
1	536365	71053	WHITE METAL LANTERN	12/1/10	3.39	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	12/1/10	2.75	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	12/1/10	2.75	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	12/1/10	2.75	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	12/1/10	2.75	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	12/1/10	2.75	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	12/1/10	2.75	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	12/1/10	2.75	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	12/1/10	2.75	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	12/1/10	2.75	
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	12/1/10	3.39	
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	12/1/10	3.39	
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	12/1/10	3.39	
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	12/1/10	3.39	



Association Rule & Preparation

```
{REGENCY TEA PLATE PINK} -> {REGENCY TEA PLATE GREEN} (conf: 0.902, supp: 0.011, lift: 61.909, conv: 10.034)
{REGENCY TEA PLATE GREEN} -> {REGENCY TEA PLATE PINK} (conf: 0.748, supp: 0.011, lift: 61.909, conv: 3.923)
{REGENCY SUGAR BOWL GREEN} -> {REGENCY MILK JUG PINK} (conf: 0.769, supp: 0.011, lift: 52.382, conv: 4.259)
{REGENCY MILK JUG PINK} -> {REGENCY SUGAR BOWL GREEN} (conf: 0.757, supp: 0.011, lift: 52.382, conv: 4.062)
{REGENCY TEA PLATE ROSES} -> {REGENCY TEA PLATE PINK} (conf: 0.601, supp: 0.011, lift: 49.700, conv: 2.474)
{REGENCY TEA PLATE PINK} -> {REGENCY TEA PLATE ROSES} (conf: 0.879, supp: 0.011, lift: 49.700, conv: 8.149)
{REGENCY TEA PLATE ROSES} -> {REGENCY TEA PLATE GREEN} (conf: 0.698, supp: 0.012, lift: 47.931, conv: 3.265)
{REGENCY TEA PLATE GREEN} -> {REGENCY TEA PLATE ROSES} (conf: 0.848, supp: 0.012, lift: 47.931, conv: 6.469)
{POPPY'S PLAYHOUSE LIVINGROOM} -> {POPPY'S PLAYHOUSE BEDROOM} (conf: 0.813, supp: 0.011, lift: 47.718, conv: 5.270)
{POPPY'S PLAYHOUSE BEDROOM} -> {POPPY'S PLAYHOUSE LIVINGROOM} (conf: 0.649, supp: 0.011, lift: 47.718, conv: 2.808)
```

- A total of 18,536 baskets were created from the invoice line items.
- The Apriori algorithm was run on these with a minimum support of 1%, resulting in 275 rules.

Bin 1	190449
Bin 3	82757
Bin 2	58781
Bin 5	6467
Bin 4	4831

- The quantity of each item in each transaction is grouped in 5 classes.
- Apriori algorithm was run to produce the rules for purchase sizes, resulting in 18 rules.

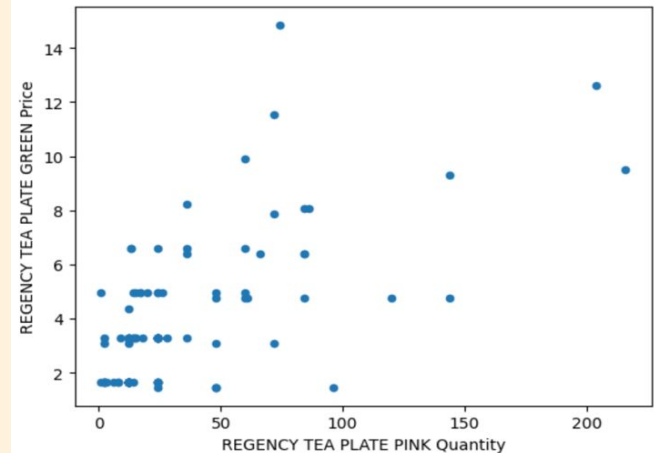
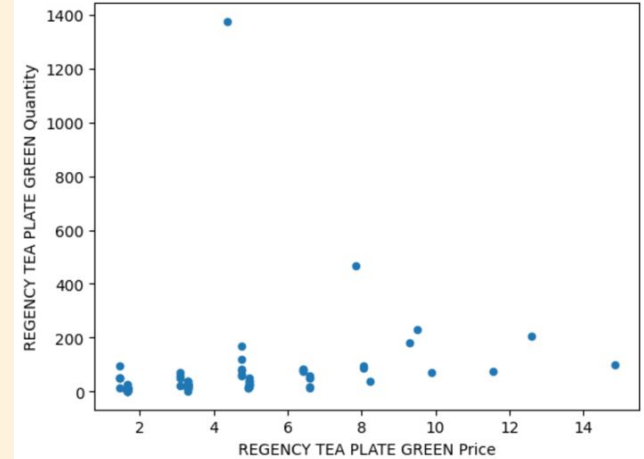
```
{PINK HAPPY BIRTHDAY BUNTING_Bin 1} -> {BLUE HAPPY BIRTHDAY BUNTING_Bin 1} (conf: 0.702, supp: 0.010, lift: 47.482, conv: 3.305)
{BLUE HAPPY BIRTHDAY BUNTING_Bin 1} -> {PINK HAPPY BIRTHDAY BUNTING_Bin 1} (conf: 0.679, supp: 0.010, lift: 47.482, conv: 3.069)
{PINK REGENCY TEACUP AND SAUCER_Bin 2} -> {GREEN REGENCY TEACUP AND SAUCER_Bin 2} (conf: 0.812, supp: 0.013, lift: 41.116, conv: 5.210)
{GREEN REGENCY TEACUP AND SAUCER_Bin 2} -> {PINK REGENCY TEACUP AND SAUCER_Bin 2} (conf: 0.637, supp: 0.013, lift: 41.116, conv: 2.709)
{ROSES REGENCY TEACUP AND SAUCER_Bin 2} -> {PINK REGENCY TEACUP AND SAUCER_Bin 2} (conf: 0.522, supp: 0.011, lift: 33.711, conv: 2.060)
{PINK REGENCY TEACUP AND SAUCER_Bin 2} -> {ROSES REGENCY TEACUP AND SAUCER_Bin 2} (conf: 0.704, supp: 0.011, lift: 33.711, conv: 3.306)
{GARDENERS KNEELING PAD KEEP CALM_Bin 1} -> {GARDENERS KNEELING PAD CUP OF TEA_Bin 1} (conf: 0.647, supp: 0.012, lift: 33.596, conv: 2.779)
{GARDENERS KNEELING PAD CUP OF TEA_Bin 1} -> {GARDENERS KNEELING PAD KEEP CALM_Bin 1} (conf: 0.647, supp: 0.012, lift: 33.596, conv: 2.779)
{ROSES REGENCY TEACUP AND SAUCER_Bin 2} -> {GREEN REGENCY TEACUP AND SAUCER_Bin 2} (conf: 0.651, supp: 0.014, lift: 32.978, conv: 2.810)
{GREEN REGENCY TEACUP AND SAUCER_Bin 2} -> {ROSES REGENCY TEACUP AND SAUCER_Bin 2} (conf: 0.689, supp: 0.014, lift: 32.978, conv: 3.143)
```

Quantity-aware Association Rules



Price-quantity Analysis

- Two strongly-related complementary items were chosen for price-quantity analysis as below:
 - Counting the quantity sold of each item per day.
 - Averaging the total price of each item per day
- Green plate quantity almost always stays at the same level with increase in price.
- With increase in green plate price, pink plate quantity tend to increase, suggesting a substitutability relationship between the products.



Techniques

Clustering Items

$$TF(Term, Item) = \frac{\text{Number Of Times Term Occurs In Item's Description}}{\text{Total Number Of Terms In Item's Description}}$$

$$IDF(Term, Item) = \frac{\text{Total Number of Items In Corpus}}{\text{Number Of Item Descriptions Containing Term}}$$

Convert collection
of items to matrix
of features

```
items = data["Description"]  
vectorizer = TfidfVectorizer()  
vec_items = vectorizer.fit_transform(items)  
print(vec_items.shape)
```

✓ 2.2s

(397925, 2031)

```
# Get the feature names (terms)
feature_names = vectorizer.get_feature_names_out()

# Print the feature names
print("Extracted Terms (Features):")
for term in feature_names:
    print(term)
```

From TF-IDF matrix:

```
Sum for all items, top 10 terms: [('bag',
np.float64(13709.915665468296)), ('set',
np.float64(12218.460272342347)), ('of',
np.float64(11367.348207858953)), ('red',
np.float64(11182.0951587633)), ('retrospot',
np.float64(10197.999731815902)), ('heart',
np.float64(9967.17316257343)), ('vintage',
np.float64(8909.511794975966)), ('design',
np.float64(8460.334863378703)), ('pink',
np.float64(7295.724298590895)), ('jumbo',
np.float64(7221.5990599103225))]
```

abstract
acapulco
account
acrylic
address
ade
adult
advent
afghan
aged
ahoy
aid
air
airline
airmail
alarm
album
all
allium
alphabet
alpine
aluminium

gin
gingerbread
gingham
giraffe
girl
girls
girly
glamorous
glamour
glass
glasses
glaze
gliders
glitter
globe
gloss
glove
gloves
glow
glowing
gnome
go
goblet
gold

winkie
wire
wireless
wise
wish
wishes
witch
with
wobbly
women
wood
wooden
woodland
wool
woolly
word
work
world
woven
wrap
wrapping
wreath
wrecking
writing
xmas
yellow
you
your
yuletide
zinc

```

num_clusters = 5
kmeans = KMeans(n_clusters=num_clusters, random_state=42)
kmeans.fit(vec_items)
results = pd.DataFrame()
results['document'] = items
results['cluster'] = kmeans.labels_

```

```

for k in range(0, 5):
    print(results[results["cluster"] == k].sample(frac=0.0002))
✓ 0.0s

```

	document	cluster
237757	3 PIECE SPACEBOY COOKIE CUTTER SET	0
366581	RETROSPOT PARTY BAG + STICKER SET	0
381302	SET OF 36 PAISLEY FLOWER DOILIES	0
287056	PACK OF 12 WOODLAND TISSUES	0
510850	SET OF 4 NAPKIN CHARMS HEARTS	0
215051	SET 12 LAVENDER BOTANICAL T-LIGHTS	0
285342	PACK OF 72 SKULL CAKE CASES	0
304398	WOODEN ROUNDERS GARDEN SET	0
419610	SET OF 60 PANTRY DESIGN CAKE CASES	0
135637	SET OF 6 T-LIGHTS EASTER CHICKS	0
113429	PACK OF 72 RETROSPOT CAKE CASES	0

document	cluster
COLOURED GLASS STAR T-LIGHT HOLDER	1
ANTIQUE SILVER T-LIGHT GLASS	1
HANGING BUTTERFLY T-LIGHT HOLDER	1
document	cluster
BLACK RECORD COVER FRAME	2
CLASSICAL ROSE SMALL VASE	2
SMALL MEDINA STAMPED METAL BOWL	2
EDWARDIAN PARASOL BLACK	2
PAINTED METAL PEARS ASSORTED	2
BAKING MOULD EASTER EGG WHITE CHOC	2
BUTTON BOX	2
JUMBO BAG RED RETROSPOT	2
PAPER BUNTING WHITE LACE	2
ROSES REGENCY TEACUP AND SAUCER	2
REVOLVER WOODEN RULER	2
CLASSIC CHROME BICYCLE BELL	2
I LOVE LONDON MINI BACKPACK	2
GINGERBREAD MAN COOKIE CUTTER	2
CLASSIC METAL BIRDCAGE PLANT HOLDER	2

document	cluster
GLITTER CHRISTMAS HEART	3
VINTAGE UNION JACK APRON	3
PAPER CHAIN KIT VINTAGE CHRISTMAS	3
VINTAGE RED TRIM ENAMEL BOWL	3
PLAYING CARDS VINTAGE DOILY	3
CHRISTMAS CRAFT LITTLE FRIENDS	3
MINI PAINT SET VINTAGE	3
document	cluster
JUMBO BAG PINK POLKADOT	4
PINK POLKADOT PLATE	4
PINK STITCHED WALL CLOCK	4
CHARLOTTE BAG PINK POLKADOT	4
JUMBO BAG PINK POLKADOT	4

Analysis Recap & Findings

- Analysis Process:
 - Data cleaning
 - Descriptive Statistics
 - Association rules, clustering items, etc.
 - Interpretations of the results

Question 1 - Can we estimate the probability of future purchases by identifying associations between products?

Question 2 - Are there meaningful insights that emerge from differences in product prices and quantities?

- ★ Descriptive statistics suggest that there is seasonality in shopper purchasing behavior
- ★ Confidence of 90.2% that invoices containing pink plates also include green ones
- ★ Green plate price and quantity have positive association. Increases in green quantity corresponds to an increases in pink plate quantity
- ★ If the price of a complementary item increases, demand for its complementary item usually decreases

Business Suggestions

- Consider coupling pink and green plates by offering deals and making these items visually available
 - Bundling discounts
 - Positioning the two items together in physical stores. Either in designated isles or in the front as a promotion
 - Integrate the two items on the homepage of the website. Using pop ups/advertisements to promote the bundling deal
- With the understanding that pink and green plates are frequently bought together rather than separate, we can make better and more accurate purchasing decisions during inventory reorders
 - Special consideration that there may be seasonality with shoppers
 - Precautionarily, inventory management needs to be reassessed periodically as the suggested marketing tactics are implemented. Assuming that we will see an increase in the pink and green plate sales

**Thank
you!**